

[Additional Exam]

VCD 05/05/2015 S.Y.B.S.C (C.S.) - COMP-I - SEM-IV - 2014-2015 - 75 - 2 $\frac{1}{2}$ HRS

- NOTE:**
- 1) Diagrams should be Neat and labeled.
 - 2) All Questions are compulsory.
 - 3) Right side indicates marks.

Q.1) Attempt any four. [20]

- A) Draw the circle with center (10,10) and radius 15 using bresenham circle.
- B) Explain arc () and rectangle () with example.
- C) Consider line AB with A=(10,20) and B=(20,12). Apply DDA line algorithm and calculate pixel on this line.
- D) Write short note on 1. Scan-conversion 2. rasterization
- E) Differentiate raster scan method and random scan method.
- F) Explain outtextxy () and circle() with example.
- G) Explain floodfill() and setfillstyle() with example.
- H) Derived steps for midpoint circle.

Q.2) Attempt any four. [20]

- A) What is inside-outside test? Explain it.
- B) Derived steps for cohen-sutherland line clipping algorithm.
- C) What is clipping ? Explain character clipping.
- D) Explain wire frame model.
- E) Describe Bezier curve with there properties.
- F) How map window port to view port. Explain it with its steps.
- G) Explain polygon clipping.
- H) Write short note on spline curve.

Q.3) Attempt any four. [20]

- A) What is rendering in computer graphics? Explain.
- B) Explain painter's algorithm.
- C) Explain z-buffer algorithm.
- D) Write a short note on 1. Shading 2. Transparency effect.
- E) Explain components of computer animation.
- F) Explain types of shadow.
- G) Describe Umbra and penumbra using shadow.
- H) Describe illumination model.

Q.4) Attempt any three. [15]

- A) Describe CRT with components.
- B) Consider line AB with A=(-1,0) and B=(-8,4). Apply DDA line algorithm and calculate pixel on this line.
- C) Write short note on 1. Computer animation 2. Morphing
- D) Explain initgraph () and detectgraph().
- E) Explain getmaxx() and getmaxy().
- F) Define image. Explain applications of computer graphics.